Number system (Assignment-1)

1. convert 0.95 to its binary number system?

$0.95 \times 2 = 1.90 \rightarrow 1$
$0.90 \times 2 = 1.80 \quad 0$
$0.80 \times 2 = 1.60 \quad 1$
$0.60 \times 2 = 1.20 \quad 1$
$0.20 \times 2 = 0.40 \quad 0$
$0.40 \times 2 = 0.80 \quad 0$
$0.80 \times 2 = 1.60 \quad 1$

$(0.11110011)_2$

2. Subtract $(1010100)_2$ from $(1010100)_2$ using 2's complement method?

1010100 - 1010100

```
1 0 1 0 1 0 0
0 1 0 1 0 1 1
1 0 1 0 1 0 0
-------------
1 1 1 1 1 1 1
```

```
1 0 1 0 1 0 0
0 1 0 1 0 1 1
        1
-------------
0 1 0 1 1 0 0
1 0 1 0 1 0 0
-------------
1 0 0 0 0 0 0 0
```

we get carry 1

so the result is +ve

ignore carry 1.

3. convert Binary code 1100 to gray code and convert gray code 11001 to binary

Bin <-> gray

```
1 <-> 1 <-> 0 <-> 0
↕     ↓     ↓     ↓
1     0     1     0
```

gray <-> Bin

```
1   1   0   0   1
↕   ↓   ↓   ↓   ↓
1   0   0   0   1
```

4. Represent the decimal (2.9) base 10 in Binary

$(2.9)_{10} \rightarrow (\quad)_2$

$2|2$
$1 \rightarrow 0$

$0.9 \times 2 = 1.8 \rightarrow 1$
$0.8 \times 2 = 1.6 \rightarrow 1$
$0.6 \times 2 = 1.2 \rightarrow 1$
$0.2 \times 2 = 0.4 \rightarrow 0$
$0.4 \times 2 = 0.8 \rightarrow 0$
$0.8 \times 2 = 1.6 \rightarrow 1$

$(10.111001)_2$

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5. Represent the Binary number $(10110.0011)_2$ in decimal, octal and hexadecimal?

Sol:- $(10110.0011)_2 \rightarrow (\ )_{10}, (\ )_8, (\ )_{16}$

$2^0\times0+2^1\times1+2^2\times1+2^3\times0+2^4\times1+2^{-1}\times0+2^{-2}\times0+2^{-3}\times1+2^{-4}\times1$

$\Rightarrow 2+4+16+0.125+0.0625$

$=(22.1875)_{10}$

$\Rightarrow (10110.0011)_2 - (\ )_8$

$010/110.0/01/1$

$(26.03)_8$

$\Rightarrow (10110.0011)_2 - (\ )_{16}$

$0001/0110.0011$

$(16.3)_{16}$

6. Find the 16's complement of $(FAB)_{16}$?

$$\begin{array}{r} FFF \\ (-)\ FAB \\ \hline 054 \\ \quad\ 1 \\ \hline 055 \end{array} \qquad X=(055)_{16}$$

7. find the value of $(34)_8-(21)_8$ using 8's complement

Sol: Find 8's complement to $(21)_8$

$$\begin{array}{r} 77 \\ 21 \\ \hline 56 \\ \quad\ 1 \\ \hline 57 \end{array}$$

$$\begin{array}{r} 1 \\ 57 \\ 34 \\ \hline 13 \end{array}$$

$8\,|\,11$ rem 3

$8\,|\,9$ rem 1

carry indicates +ve sign

8) $(11001)_2-(11011010)_2$

Both the bit are not same no of count so we use 2's complement method for subtraction

```
 11011010
 00100101
        1
 --------
 00100110
 00011001
 --------
 01001111111
  11000000
         1
 --------
 11000001
```

there is no carry
repate 2's complement

$(11001)_2 - (11011010)_2 = (11000001)_2$.

9. $(DEAD)_{16} + (beef)_{16} = ()_8$

solr first convert hexadecimal to decimal

D = 13, E = 14, A = 10

$13\times16^0 + 10\times16^1 + 14\times16^2 + 13\times16^3$

$= 53248 + 3584 + 160 + 13 = 56905$

$(BEEF)_{16}$

$15\times16^0 + 14\times16^1 + 14\times16^2 + 11\times16^3$

$= 45056 + 3584 + 224 + 15 = 48879$

Add $56905 + 48879 = (105784)_{10}$ (decimal)

convert into octal

```
8 | 105784
8 | 13223 -> 0
8 | 1652  -> 7
8 | 206   -> 4
8 | 25    -> 6
8 | 3     -> 1
```

$= (316470)_8$

10. Convert the binary $(11010110)_2$ to Gray

```
1 <-> 1 <-> 0 (-> 1 <-> 0 (-> 1 <-> 1 <-> 0
↓     ↓     ↓     ↓     ↓     ↓     ↓     ↓
1     0     1     1     1     1     0     1
```

$= (10111101)_2$

11. find the decimal for BCD representation of $(1101\,1010\,1100)_{BCD}$?

Sol: given binary group in decimal

1101 → 13

1010 → 10

1100 → 12

In standard BCD the binary values 1010, 1100, and 1101 are invalid. BCD only use the binary represention of 0 through 9.

∴ The given number is not a valid BCD number.

12. Represent the decimal number $(48)_{10}$ using following weighted binary code 5311

Sol: $(48)_{10}$

4 → 0110

8 → 1100

$(49)_{10} = (0110\,1100)$

13. $(3611349/147)_{10} = (57767)_x$ find base value

Sol: $\frac{3611349}{147} = 24,567$

give 57767

we can observe that all the digits is b/w 0 to 7

There fore the base must be at least 8

To verify

let convert (57767) to decimal

$5\times8^4 + 7\times8^3 + 7\times8^2 + 6\times8^1 + 7\times8^0$

$= 20480 + 3584 + 448 + 48 + 7 = 24567$

$\therefore (57767)_8$

14. $\frac{(30165732)_8}{(147)_{10}}$ convert into hexadecimal (16)?

Sol' first NR value that convert octal to decimal.

$(30165732)_8 = (\quad)_{10}$

$2\times8^0+3\times8^1+3\times8^2+7\times8^3+5\times8^4+6\times8^5+1\times8^6+0\times8^7$
$+3\times8^8$

$= 50331648+0+262144+196608+20480+3584$
$+192+8+2$
$=(50814666)_{10}$

now $\frac{50814666}{147} = 345678$

Convert in hexdicimal $(345678)_{10} = (\quad)_{16}$

16 | 345678
16 | 21604 → 14 (E)
16 | 1350 → 4
16 | 84 → 6
5 → 4

$(5464E)_{16}$

15. $(567)_{10}$ is encoded in various 4 bit codes.

1. BCD (8421) → 0101 0110 0111
2. 5311 code → 1000 1001 1011
3. Excess-3 code → 1000 1001 1010
4. 2421 code → 0101 0110 0111
5. Grey code → 0111 0101 0100.

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